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CSC 177 - Assignment 2 Report

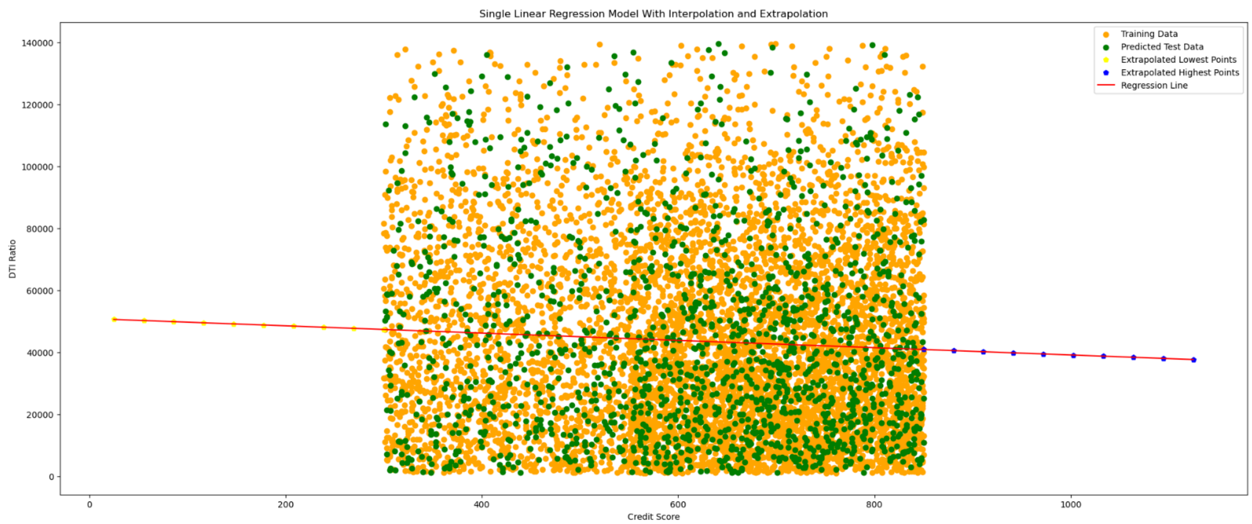
Pre-Part A - Preprocessing for the New Dataset

We chose to go ahead with a different dataset for this assignment. This dataset dealt with the approval odds of a loan of the applicant. With all the preprocessing done such as; counting and dropping any missing rows of data, removing the duplicate rows, removing outliers using the interquartile range method, converting the categorical column to numerical and plotting, we plotted the data. We also had to balance the dataset because as it was, about 80% of the data was leaning to the "reject" outcome which would not only heavily skew the data plot but also result in a poor model training. We did this by downsampling the "rejected" states to match the same number of dataset for the "approved" one. As a result, we got about 50-50 for both; "reject" and "approved" rows. We then split the dataset into 20% training and 80% testing model using the built-in library, "train_test_split.

Part A - Regression Analysis (Single)

For the single linear regression model, we chose the "Credit_Score" column as the input for "x" values and "DTI_Score" for the observed values for "y" training the model. We then tested the x and y training model with their test models. These would be our interpolated data with the predicted y values from the model as no predicted y values would escape the minimum and maximum values of the observed data. Next, We tried to handle the extrapolation by first getting the minimum and maximum values of the training data. This was done so we could generate extrapolated points which would be below this minimum value as well as values which would be greater than the maximum. We attained this using *linspace.* We then combined the final input x as these two extrapolated points (as a list) as well as our original x inputs and then predicted the y values. We sorted the data before plotting to get a smooth regression line.

Result

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The Root Mean Square Error = 31771.7144
The R-squared = 0.0042
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Interpretation

For the interpolation data, we see a good scatter of the training and the test data's predicted values as they all fell in the range of the minimum and maximum “x” values with heavier clumps below the 50000 ratio. We can see an almost horizontal regression line which may signify that the linear relationship between the DTI_Score and the Credit_Score is a weak one. This may also be because of the dense scatter of the data where the line maybe capturing only little variance. Since there doesn't seem to be a strong linear relationship, we can interpret that as the credit score of one increases, it may not necessarily increase or decrease the DTI ratio of the applicant. We verified this by calculating the R-Squared value between these two variables and saw that it came to 0.0042 which is almost nil.

Using extrapolation, the same linear regression from the interpolation persists only with the extension of this line beyond the training and tested data values on both ends. This now helps us see the full regression line in that it is tipping in a downward slope. This extension will continue because we do not have any training data in those extreme regions so the model is holding a pure guess game that this trend will continue forever.

In both the cases, the prediction denotes little to no correlation between the Credit Score and the DTI ratio.

Part A - Regression Analysis (Multiple)

For the multiple linear regression model, we chose the “Credit_Score” column as the target value. We then interpolated the data by choosing the remaining columns as the features and passed it for training and testing the model. For extrapolation, we did the same as above; however, to generate the values out of the range, we would require the mean values of the feature columns. This was needed for generating the values below and above the minimum and maximum training values, respectfully. We then tested the training model with their test models by supplying these values in bulk to get the predicted “y” values. We attained this again by using *linspace.* We then combined the final input “x” as these two extrapolated points (as a list) as well as our original “x” inputs and then predicted the “y” values. We sorted the data before plotting to get a smooth regression line.

Result


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The Root Mean Square Error = 123.5643
The R-squared = 0.3204
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Interpretation

The interpolation was done the similar way to the single regression. We can see the predicted test data and the training data have overlaps which tells us that the model's predicted values were similar if not same to the observed data used for training it. We can see the downward slope of the regression line in the middle once again which suggests a slight negative relationship of income to the credit score. This makes sense logically that a higher income does not necessarily mean a better credit score but almost frequently, the lower the income, the lower the credit score of an individual. The calculated R-squared score also explains this correlation. It denotes that the model can explain 32% of the variance while other features are held constant and income alone is not the predominant feature in determining the credit score.

Extrapolation was done in a similar way too. Since our model was linear, the regression line was extended beyond the minimum and maximum data points. While holding other features constant using their mean values, the regression line allowed us to clearly see the trend in the regression line.

All in all, the overall results were pretty similar for both, the single as well as the multiple line regression with a noticeable change in the R-squared value between the both.